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RECIPROCALS OF PARTITION POLYNOMIALS

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Abstract. Ballantine–Beck–Feigon–Maurischat introduced the subsum polynomial

$$\mathrm{sp}(\lambda, x) := \prod_i (1 + x^{\lambda_i})$$

attached to an integer partition λ, and studied rational functions obtained by summing reciprocals of these polynomials over natural classes of partitions. They posed ten conjectures which naturally divide into coprimality and divisibility questions, special-value and recurrence formulas, and coefficient-shape problems. We prove all of the conjectures in the first two families: the ordinary and binary coprimality/divisibility conjectures, and the odd and ternary special-value/recurrence conjectures. AxiomProver autonomously produced Lean/mathlib formalizations and machine-checkable proofs of these six conjectures, and also discovered the counterexample showing that the binary log-concavity conjecture is false as stated.

1. Introduction

1.1. **Background.** The study of integer partitions is a classical subject in number theory and combinatorics; we follow the standard notation and terminology described, for example, in Andrews' book [1]. A *partition* of a nonnegative integer n is a finite nonincreasing sequence

$$\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_k)$$

of positive integers whose sum is n. We write $\lambda \vdash n$. The empty sequence is the unique partition of 0. The length of λ is $\ell(\lambda) := k$, and $m_\lambda(i)$ denotes the multiplicity of the part i in λ. In particular, we have

$$\sum_{i \geq 1} i\, m_\lambda(i) = n \qquad \text{and} \qquad \ell(\lambda) = \sum_{i \geq 1} m_\lambda(i).$$

Ballantine–Beck–Feigon–Maurischat recently introduced [2] a family of interesting rational functions attached to partitions. We now recall their construction in the notation used throughout this paper. For a partition λ, its *subsum polynomial* is

$$\mathrm{sp}(\lambda, x) := \prod_{j=1}^{k} (1 + x^{\lambda_j}).$$

Equivalently, we have

$$\mathrm{sp}(\lambda, x) = \prod_{i \geq 1} (1 + x^i)^{m_\lambda(i)}.$$

This polynomial is the generating function for submultisets of the multiset of parts of λ. The reciprocal sum over ordinary partitions is

$$\mathrm{sr}(n, x) := \sum_{\lambda \vdash n} \frac{1}{\mathrm{sp}(\lambda, x)}. \tag{1}$$

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1.1.1. *The numerator and denominator of* $\mathrm{sr}(n,x)$. A common denominator for (1) is

$$\mathrm{den}^*(n,x) \coloneqq \prod_{i\geq 1}(1+x^i)^{\lfloor n/i\rfloor},$$

and the corresponding numerator is

$$\mathrm{num}^*(n,x) \coloneqq \sum_{\lambda\vdash n} h_\lambda(x),$$

where

$$h_\lambda(x) \coloneqq \prod_{i\geq 1}(1+x^i)^{\lfloor n/i\rfloor - m_\lambda(i)}.$$

(The products are finite in effect, since $\lfloor n/i\rfloor = 0$ for $i > n$. The exponents of $h_\lambda(x)$ are nonnegative because a partition of n can contain at most $\lfloor n/i\rfloor$ parts equal to i.) In other words,

$$\mathrm{sr}(n,x) = \frac{\mathrm{num}^*(n,x)}{\mathrm{den}^*(n,x)}. \tag{2}$$

However the quotient (2) need not be in lowest terms. The first cancellation considered by Ballantine–Beck–Feigon–Maurischat removes the common divisor of the individual summands. Define

$$G(n,x) \coloneqq \gcd\{h_\lambda(x) : \lambda\vdash n\},$$

and then define

$$\mathrm{num}(n,x) \coloneqq \frac{\mathrm{num}^*(n,x)}{G(n,x)} \qquad \text{and} \qquad \mathrm{den}(n,x) \coloneqq \frac{\mathrm{den}^*(n,x)}{G(n,x)}. \tag{3}$$

Therefore, we have

$$\mathrm{sr}(n,x) = \frac{\mathrm{num}(n,x)}{\mathrm{den}(n,x)}. \tag{4}$$

We use the empty-partition conventions $G(0,x)=1$, $h_\emptyset(x)=1$, $\mathrm{num}(0,x)=1$.

We stress that $G(n,x)$ is *not* defined to be $\gcd(\mathrm{num}^*(n,x),\mathrm{den}^*(n,x))$. It is the common factor of the summands $h_\lambda(x)$. Therefore, it is conceivable *a priori* that $\mathrm{num}(n,x)$ and $\mathrm{den}(n,x)$ are not relatively prime as polynomials in $\mathbb{Z}[x]$.

1.1.2. *Example for* $n=4$. Here we write out the example $n=4$ in full. The five partitions of 4 are

$$(4), \qquad (3,1), \qquad (2,2), \qquad (2,1,1), \qquad (1,1,1,1),$$

so their subsum polynomials are

$$\begin{aligned}
\lambda=(4) &\implies \mathrm{sp}(\lambda,x) = 1+x^4\\
\lambda=(3,1) &\implies \mathrm{sp}(\lambda,x) = (1+x^3)(1+x)\\
\lambda=(2,2) &\implies \mathrm{sp}(\lambda,x) = (1+x^2)^2\\
\lambda=(2,1,1) &\implies \mathrm{sp}(\lambda,x) = (1+x^2)(1+x)\\
\lambda=(1,1,1,1) &\implies \mathrm{sp}(\lambda,x) = (1+x)^4.
\end{aligned}$$

Therefore, we have

$$\mathrm{sr}(4,x) = \frac{1}{1+x^4} + \frac{1}{(1+x^3)(1+x)} + \frac{1}{(1+x^2)^2} + \frac{1}{(1+x^2)(1+x)^2} + \frac{1}{(1+x)^4}.$$

The unreduced common denominator is

$$\mathrm{den}^*(4,x) = (1+x)^4(1+x^2)^2(1+x^3)(1+x^4).$$

The five terms $h_\lambda(x) = \mathrm{den}^*(4,x)/\operatorname{sp}(\lambda,x)$ are

$$\begin{aligned}
\lambda = (4) &\implies h_\lambda(x) = (1+x)^4(1+x^2)^2(1+x^3) \\
\lambda = (3,1) &\implies h_\lambda(x) = (1+x)^3(1+x^2)^2 \qquad (1+x^4) \\
\lambda = (2,2) &\implies h_\lambda(x) = (1+x)^4 \qquad (1+x^3)(1+x^4) \\
\lambda = (2,1,1) &\implies h_\lambda(x) = (1+x)^2(1+x^2)\ (1+x^3)(1+x^4) \\
\lambda = (1,1,1,1) &\implies h_\lambda(x) = \qquad (1+x^2)^2(1+x^3)(1+x^4)
\end{aligned}$$

and $\mathrm{num}^*(4,x) = \sum_{h\vdash 4} h_\lambda(x)$. Since $1+x$ divides $1+x^3$, each of these polynomials is divisible by $1+x$, and in fact this is the greatest common factor:

$$G(4,x) = \gcd\{h_\lambda(x)\ : \lambda \vdash 4\} = 1+x.$$

After dividing by this common factor, we obtain

$$\mathrm{den}(4,x) = \frac{\mathrm{den}^*(4,x)}{1+x} = (1+x)^3(1+x^2)^2(1+x^3)(1+x^4),$$

and

$$\begin{aligned}
\mathrm{num}(4,x) &= \frac{\mathrm{num}^*(4,x)}{1+x} \\
&= 5x^{10} + 8x^9 + 15x^8 + 14x^7 + 24x^6 + 20x^5 + 24x^4 + 14x^3 + 15x^2 + 8x + 5.
\end{aligned}$$

In this case $\mathrm{sr}(4,x) = \frac{\mathrm{num}(4,x)}{\mathrm{den}(4,x)}$ is already in lowest terms.

1.1.3. *Restricted classes of partitions.* We refer to the partitions of n as *ordinary partitions*. We also use the following special classes: $\mathcal{O}(n)$ is the set of odd partitions of n, meaning partitions all of whose parts are odd; $\mathcal{B}(n)$ is the set of binary partitions of n, meaning partitions all of whose parts are powers of 2; and $\mathcal{T}(n)$ is the set of ternary partitions of n, meaning partitions all of whose parts are powers of 3.

The same notation will be used for the restricted classes of partitions. If $\mathcal{C}$ is one of $\mathcal{O}, \mathcal{B}, \mathcal{T}$, let $A_\mathcal{C}$ be its set of allowed parts:

$$A_\mathcal{O} \coloneqq \{1,3,5,\ldots\}, \qquad A_\mathcal{B} \coloneqq \{1,2,4,8,\ldots\}, \qquad A_\mathcal{T} \coloneqq \{1,3,9,27,\ldots\}.$$

For $\lambda \in \mathcal{C}(n)$, define

$$\begin{aligned}
h_{\mathcal{C},\lambda}(x) &\coloneqq \prod_{i\in A_\mathcal{C}} (1+x^i)^{\lfloor n/i\rfloor - m_\lambda(i)} \\
G_\mathcal{C}(n,x) &\coloneqq \gcd_{\lambda\in\mathcal{C}(n)} h_{\mathcal{C},\lambda}(x),
\end{aligned}$$

and

$$\mathrm{num}_\mathcal{C}(n,x) \coloneqq \frac{1}{G_\mathcal{C}(n,x)} \sum_{\lambda\in\mathcal{C}(n)} h_{\mathcal{C},\lambda}(x) \qquad \text{and} \qquad \mathrm{den}_\mathcal{C}(n,x) \coloneqq \frac{\prod_{i\in A_\mathcal{C}}(1+x^i)^{\lfloor n/i\rfloor}}{G_\mathcal{C}(n,x)}.$$

1.2. **Ballantine–Beck–Feigon–Maurischat's Ten Conjectures.** In the recent paper [2], Ballantine–Beck–Feigon–Maurischat make ten explicit conjectures about these reciprocals of partition polynomials. We list them here in the notation fixed above, so that the statements proved later can be compared directly with the original conjectures.

1.2.1. *Ordinary partitions.* The first two conjectures concern the ordinary numerator and ask whether it is irreducible, and then whether (4) is already in lowest terms.

Conjecture 1. *For $n \geq 1$, the polynomial $\mathrm{num}(n,x)$ is irreducible over $\mathbb{Z}$.*

Conjecture 2. *For $n \geq 1$, we have $\gcd(\mathrm{num}(n,x), \mathrm{den}(n,x)) = 1$.*

The following two conjectures concern coefficient-shape properties of the ordinary numerator and denominator. Recall that a finite sequence is *unimodal* if it weakly increases up to some point and then weakly decreases; and a finite sequence (a_i) of nonnegative numbers is *log-concave* if $a_i^2 \geq a_{i-1}a_{i+1}$ for every internal index i.

Conjecture 3. *Write $\mathrm{num}(n,x) = \mathrm{num}_0(n,x) + \mathrm{num}_1(n,x)$, where num_0 contains the even powers of x and num_1 contains the odd powers of x. Then $\mathrm{num}_0(n,x)$ is unimodal.*

Conjecture 4. *The sequence of coefficients of $\mathrm{den}(n,x)$ is log-concave except for $n = 3, 5, 6, 7$.*

1.2.2. *Binary partitions.* The next three conjectures concern binary partitions.

Conjecture 5. *For the binary numerator and denominator, we have*

$$\gcd(\mathrm{num}_{\mathcal{B}}(n,x), \mathrm{den}_{\mathcal{B}}(n,x)) = 1 \qquad (n \geq 1).$$

Conjecture 6. *For $n \geq 2$, the binary numerator $\mathrm{num}_{\mathcal{B}}(n,x)$ is unimodal.*

Conjecture 7. *For $n \geq 2$ and every $2^s \leq n$, we have*

$$(1 + x^{2^s}) \nmid \mathrm{num}_{\mathcal{B}}(n,x).$$

Remark. It is noted in [2] that Conjecture 5 and Conjecture 7 are equivalent.

Remark. The formulation of Conjecture 6 in [2] additionally asserts that $\mathrm{num}_{\mathcal{B}}(n,x)$ is log-concave. However, this doesn't appear to be true for $n = 4$, where

$$\mathrm{num}_{\mathcal{B}}(4,x) = 4 + 10x + 18x^2 + 18x^3 + 20x^4 + 18x^5 + 18x^6 + 10x^7 + 4x^8.$$

The coefficient sequence is unimodal, but it is not log-concave, since $18^2 < 18 \cdot 20$.

1.2.3. *Odd and ternary partitions.* The remaining conjectures[1] treated here involve odd and ternary partitions. We write $v_p(M)$ for the exponent of the prime p in a positive integer M.

Conjecture 8. *If $o(m)$ denotes the largest odd divisor of m, then $\mathrm{num}_{\mathcal{O}}(n,-1) = o(n!)$ for $n \geq 1$.*

Conjecture 9. *Let $s(n) := \mathrm{num}_{\mathcal{T}}(n,-1)$ for $n \geq 1$. Then $s(1) = s(2) = 1$, and $s(3n) = s(3n+1) = s(3n+2) = 3^{v_3((3n)!)}$ for $n \geq 1$.*

Conjecture 10. *For $n \geq 0$, let $t(n) := \mathrm{num}_{\mathcal{T}}(n,1)$, with the convention $t(0) := 1$. Then $t(3n) = t(3n+1) = t(3n+2)$ for $n \geq 0$, and*

$$t(n) = \frac{t(3n) - t(3n-2)}{2^{2n}}$$

for $n \geq 1$. In particular, $t(1) = t(2) = 1$.

[1] The formulation of Conjecture 10 printed in [2] reads $t(n) = \Delta(t(3n-2)/2^{2n})$. If read literally as applying the finite-difference operator to the sequence $\frac{t(3n-2)}{2^{2n}}$, then the statement is false, since, e.g., $\frac{t(4)}{2^4} - \frac{t(1)}{2^2} = \frac{5}{16} - \frac{1}{4} = \frac{1}{16} \neq t(1) = 1$. We believe that the statement given in Conjecture 10 is the intended reading (note we have used $t(3n+1) = t(3n)$).

1.3. **Results.** In this note we prove six of the ten conjectures using elementary number theory and facts about cyclotomic polynomials. The precise results are as follows.

Theorem 1 (Conjecture 2)**.** *For every $n \geq 1$, we have*

$$\gcd(\mathrm{num}(n,x), \mathrm{den}(n,x)) = 1.$$

Theorem 2 (Conjectures 5 and 7)**.** *For every $n \geq 2$ and every $2^s \leq n$, we have*

$$1 + x^{2^s} \nmid \mathrm{num}_{\mathcal{B}}(n,x).$$

Consequently, for every $n \geq 1$, we have

$$\gcd(\mathrm{num}_{\mathcal{B}}(n,x), \mathrm{den}_{\mathcal{B}}(n,x)) = 1.$$

Theorem 3 (Conjecture 8)**.** *For every $n \geq 1$, we have*

$$\mathrm{num}_{\mathcal{O}}(n,-1) = o(n!),$$

where $o(M)$ denotes the largest odd divisor of M.

Theorem 4 (Conjecture 9)**.** *For every $n \geq 1$, we have*

$$\mathrm{num}_{\mathcal{T}}(n,-1) = 3^{v_3(n!)}.$$

In particular, for every $n \geq 1$, we have

$$\mathrm{num}_{\mathcal{T}}(3n,-1) = \mathrm{num}_{\mathcal{T}}(3n+1,-1) = \mathrm{num}_{\mathcal{T}}(3n+2,-1) = 3^{v_3((3n)!)}.$$

Theorem 5 (Conjecture 10)**.** *Let $t(n) \coloneqq \mathrm{num}_{\mathcal{T}}(n,1)$ and put $t(0) \coloneqq 1$. Then*

$$t(3n) = t(3n+1) = t(3n+2) \text{ for } n \geq 0,$$
$$t(3n) - t(3n-2) = 2^{2n} t(n) \text{ for } n \geq 1.$$

The results above resolve the conjectures of Ballantine–Beck–Feigon–Maurischat that concern coprimality, divisibility, and special values. For clarity, we summarize below the status of all ten conjectures in their paper. The remaining unresolved cases are the conjectures concerning irreducibility and coefficient-shape phenomena; in the binary case, the log-concavity assertion is false as stated.

#	**Topic**	**Status in this paper**
1	Irreducibility of $\mathrm{num}(n,x)$ over $\mathbb{Z}$	Open
2	$\gcd(\mathrm{num}(n,x), \mathrm{den}(n,x)) = 1$	Proved
3	Unimodality of the even part $\mathrm{num}_{\mathcal{O}}(n,x)$	Open
4	Log-concavity of $\mathrm{den}(n,x)$ for $n \notin \{3,5,6,7\}$	Open
5	$\gcd(\mathrm{num}_{\mathcal{B}}(n,x), \mathrm{den}_{\mathcal{B}}(n,x)) = 1$	Proved
6	Unimodality and log-concavity of $\mathrm{num}_{\mathcal{B}}(n,x)$	Disproved log-concavity
7	$(1 + x^{2^i}) \nmid \mathrm{num}_{\mathcal{B}}(n,x)$ for $2^i \leq n$	Proved
8	$\mathrm{num}_{\mathcal{O}}(n,-1) = o(n!)$ for odd partitions	Proved
9	Evaluation formula for $\mathrm{num}_{\mathcal{T}}(n,-1)$	Proved
10	Recurrence formula for $\mathrm{num}_{\mathcal{T}}(n,1)$	Proved in corrected form

TABLE 1. Status of the BBFM conjectures addressed in this paper.

The paper is organized as follows. In Section 2 we record the elementary cyclotomic facts that will be used throughout the proofs. In Section 3, we prove Conjecture 2 for ordinary partitions by showing that the reduced numerator does not vanish at any root of unity that can occur as a zero of the reduced denominator. The key step is a reduction to a smaller remainder problem in which all surviving summands lie on a single open ray in the complex plane. In Section 4, we adapt the same idea to binary partitions, proving the nondivisibility statement of Conjecture 7 and hence the equivalent coprimality statement of Conjecture 5. In Section 5, we prove the evaluation in Conjecture 8 for odd partitions by specializing at $x = -1$, where only the all-ones partition survives after cancellation. In Section 6, we treat ternary partitions: the specialization at $x = -1$ gives Conjecture 9, while the specialization at $x = 1$ leads to a simple generating-function identity and proves the natural finite-difference form of Conjecture 10. The appendix documents formal proofs of the theorems we proved, and provides links to the relevant files for interested readers. (Mathematicians not interested in automated theorem proving can thus safely ignore the appendix.)

2. Elementary cyclotomic facts

Let Φ_m denote the mth cyclotomic polynomial, irreducible over $\mathbb{Z}$. We record two elementary facts about Φ_m. They are included to make the proof accessible to readers who have seen cyclotomic polynomials but may not have used them in this way. The first lemma identifies exactly which factors $1 + x^i$ contain a given cyclotomic divisor.

Lemma 1. *Let $d, i \geq 1$. Then $\Phi_{2d}(x)$ divides $1 + x^i$ if and only if $i = dj$ for some odd integer $j \geq 1$. When this occurs, $\Phi_{2d}(x)$ divides $1 + x^i$ exactly once.*

Proof. Let $\zeta_{2d} := e^{\pi \mathrm{i}/d}$. Then ζ_{2d} is a primitive $2d$-th root of unity and $\zeta_{2d}^d = -1$. The polynomial Φ_{2d} divides $1 + x^i$ exactly when $\zeta_{2d}^i = -1$. This happens exactly when

$$i \equiv d \pmod{2d},$$

which is the same as saying $i = dj$ for an odd integer $j \geq 1$. The second part follows by noting $1 + x^i$ has no double roots. □

The second lemma records the values at 1 and -1 that will be needed in the odd and ternary specializations.

Lemma 2. *For $m > 1$, we have*

$$\Phi_m(1) = \begin{cases} p, & m = p^a \text{ is a power of a prime } p, \\ 1, & m \text{ is not a prime power.} \end{cases}$$

If $m > 1$ is odd, then $\Phi_{2m}(-1) = \Phi_m(1)$. In particular, for $a \geq 1$, we have

$$\Phi_{2\cdot 3^a}(1) = 1, \qquad \Phi_{2\cdot 3^a}(-1) = 3.$$

Proof. Let μ denote the classical Möbius function. The standard product formula

$$\Phi_m(x) = \prod_{e \mid m} (x^e - 1)^{\mu(m/e)}$$

implies, after taking the limit as $x \to 1$, that

$$\Phi_m(1) = \prod_{e \mid m} e^{\mu(m/e)}.$$

For a fixed prime p, let v_p denote the usual p-adic valuation. The exponent of p in this product is

$$\sum_{e \mid m} \mu(m/e) v_p(e).$$

Writing $e = m/f$, this becomes

$$\sum_{f|m} \mu(f)(v_p(m) - v_p(f)).$$

Since $m > 1$, $\sum_{f|m} \mu(f) = 0$. Therefore, the exponent is $-\sum_{f|m} \mu(f)v_p(f)$. This is 1 if m is a power of p, and is 0 otherwise. This proves the formula for $\Phi_m(1)$.

If m is odd, then $\Phi_{2m}(x) = \Phi_m(-x)$. Therefore $\Phi_{2m}(-1) = \Phi_m(1)$. Finally, $\Phi_{2\cdot 3^a}(x) = \Phi_{3^a}(-x)$, so $\Phi_{2\cdot 3^a}(-1) = \Phi_{3^a}(1) = 3$, while

$$\Phi_{2\cdot 3^a}(1) = \Phi_{3^a}(-1) = 1. \qquad \square$$

3. Ordinary partitions: proof of Conjecture 2

Throughout this section, $m_\lambda(i)$, h_λ, and $G(n, x)$ have the meanings fixed in the introduction. We do not need a closed formula for the reduced denominator. In what follows it will be useful to write

$$\mathrm{num}(n, x) = \sum_{\lambda \vdash n} q_\lambda(x),$$

where

$$q_\lambda(x) \coloneqq \frac{h_\lambda(x)}{G(n, x)}.$$

We now prove Theorem 1. Since $\mathrm{den}(n, x) = \mathrm{den}^*(n, x)/G(n, x)$ divides $\mathrm{den}^*(n, x)$, every irreducible factor of $\mathrm{den}(n, x)$ is a cyclotomic factor of some $1 + x^i$ with $1 \le i \le n$. Therefore, every irreducible factor of $\mathrm{den}(n, x)$ has the form $\Phi_{2d}(x)$ for some $1 \le d \le n$. It is enough to prove that $\mathrm{num}(n, x)$ does not vanish at a primitive $2d$-th root of unity for any such d. The first lemma computes how much of each possible cyclotomic factor is removed by the common divisor $G(n, x)$.

Lemma 3. *Fix $n, d \ge 1$, and put $a \coloneqq \lfloor n/d \rfloor$. The exponent of $\Phi_{2d}(x)$ in $G(n, x)$ is*

$$\sum_{\substack{j>1 \\ j \text{ odd}}} \left\lfloor \frac{n}{dj} \right\rfloor.$$

Consequently the exponent of $\Phi_{2d}(x)$ in $q_\lambda(x) = h_\lambda(x)/G(n, x)$ is

$$a - \sum_{\substack{j\ge 1 \\ j \text{ odd}}} m_\lambda(dj).$$

Proof. By Lemma 1, the exponent of $\Phi_{2d}(x)$ in $h_\lambda(x)$ is

$$\sum_{\substack{j\ge 1 \\ j \text{ odd}}} \left(\left\lfloor \frac{n}{dj} \right\rfloor - m_\lambda(dj) \right).$$

The exponent in $G(n, x)$ is the minimum of this quantity over all partitions $\lambda \vdash n$.

The multiplicity sum $\sum_{j\ge 1,\, j \text{ odd}} m_\lambda(dj)$ is at most $\lfloor n/d \rfloor$, because the counted parts have total size at least $d \sum_{j\ge 1,\, j \text{ odd}} m_\lambda(dj)$. This upper bound is attained by a partition containing exactly $a = \lfloor n/d \rfloor$ parts equal to d, with the remaining $n - ad$ filled by ones. Hence the minimum exponent is obtained by subtracting a from the full floor sum:

$$\sum_{\substack{j\ge 1 \\ j \text{ odd}}} \left\lfloor \frac{n}{dj} \right\rfloor - a = \sum_{\substack{j>1 \\ j \text{ odd}}} \left\lfloor \frac{n}{dj} \right\rfloor.$$

Subtracting this minimum from the exponent in h_λ gives the exponent in q_λ. $\square$

The next lemma identifies the only summands that can survive after evaluating at a primitive $2d$-th root of unity. It reduces the problem from n to the remainder of n modulo d.

Lemma 4 (Remainder reduction)**.** *Let $n, d \geq 1$, and write*

$$n = ad + r, \qquad a \coloneqq \lfloor n/d \rfloor, \qquad 0 \leq r < d.$$

Then we have

$$\operatorname{num}(n, \zeta_{2d}) = C_{n,d} \operatorname{num}(r, \zeta_{2d})$$

for some nonzero complex number $C_{n,d}$. In particular, we have

$$\operatorname{num}(n, \zeta_{2d}) \neq 0 \quad \iff \quad \operatorname{num}(r, \zeta_{2d}) \neq 0.$$

Proof. We first start by analyzing which terms of the sum

$$\operatorname{num}(n, \zeta_{2d}) = \sum_{\lambda \vdash n} q_\lambda(\zeta_{2d})$$

are nonzero. By Lemma 3, the exponent of Φ_{2d} in the reduced summand q_λ is

$$a - \sum_{\substack{j \geq 1 \\ j \text{ odd}}} m_\lambda(dj).$$

At $x = \zeta_{2d}$, a summand survives if and only if this exponent is zero, i.e.,

$$\sum_{\substack{j \geq 1 \\ j \text{ odd}}} m_\lambda(dj) = a.$$

If any counted part dj has $j > 1$, then one of the a counted parts has size at least $3d$, so the counted parts alone have total size at least $(a-1)d + 3d = (a+2)d > ad + r = n$. This is impossible. We conclude the surviving partitions are exactly

$$\lambda = (d^a, \nu), \qquad \nu \vdash r,$$

with the empty partition allowed when $r = 0$. In other words,

$$\operatorname{num}(n, \zeta_{2d}) = \sum_{\nu \vdash r} q_{(d^a, \nu)}(\zeta_{2d}). \tag{5}$$

Next we rewrite $q_{(d^a,\nu)}$ in terms of q_ν for $\nu \vdash r$. Comparison of exponents gives the identity, in the field of rational functions,

$$q_{(d^a,\nu)}(x) = F_{n,d}(x) q_\nu(x),$$

where

$$F_{n,d}(x) \coloneqq \frac{G(r,x)}{G(n,x)} \prod_{i \neq d} (1 + x^i)^{\lfloor n/i \rfloor - \lfloor r/i \rfloor}.$$

This factor is independent of ν, so (5) implies

$$\operatorname{num}(n, \zeta_{2d}) = C_{n,d} \operatorname{num}(r, \zeta_{2d}) \qquad \text{where } C_{n,d} \coloneqq F_{n,d}(\zeta_{2d}).$$

It remains only to verify that $C_{n,d} \neq 0$.

A factor $1 + x^i$ can vanish at ζ_{2d} only through the cyclotomic factor Φ_{2d}. Since $r < d$, no factor occurring in $\operatorname{den}^*(r,x)$, and hence no factor of $G(r,x)$, is divisible by Φ_{2d}. Thus $G(r, \zeta_{2d}) \neq 0$. The product

$$\prod_{i \neq d} (1 + x^i)^{\lfloor n/i \rfloor - \lfloor r/i \rfloor}$$

has Φ_{2d}-exponent

$$\sum_{\substack{j>1\\ j \text{ odd}}} \left(\left\lfloor \frac{n}{dj} \right\rfloor - \left\lfloor \frac{r}{dj} \right\rfloor \right) = \sum_{\substack{j>1\\ j \text{ odd}}} \left\lfloor \frac{n}{dj} \right\rfloor,$$

because $r < d$. By Lemma 3, this is exactly the Φ_{2d}-exponent of $G(n,x)$. After cancelling these equal powers of Φ_{2d}, every remaining numerator and denominator factor is nonzero at ζ_{2d}. □

The reduction leaves the case $0 \le r < d$. The following same-ray argument shows that the remaining terms cannot cancel.

Lemma 5. *Let $d \ge 1$ and $0 \le r < d$. Then*

$$\operatorname{num}(r, \zeta_{2d}) \neq 0.$$

Proof. If $r = 0$, then $\operatorname{num}(0,x) = 1$. Assume $1 \le r < d$. For $1 \le i \le r$,

$$1 + \zeta_{2d}^{i} = 1 + e^{\pi \mathrm{i} i/d} = 2\cos\left(\frac{\pi i}{2d}\right) e^{\pi \mathrm{i} i/(2d)}.$$

The cosine is strictly positive because $0 < i < d$. Thus each factor $1 + \zeta_{2d}^{i}$ has positive length and argument $\pi i/(2d)$.

For a partition $\nu \vdash r$, the value $h_\nu(\zeta_{2d})$ has argument

$$\frac{\pi}{2d} \sum_{i=1}^{r} i(\lfloor r/i \rfloor - m_\nu(i)).$$

But $\sum_i i m_\nu(i) = r$, so this argument equals

$$\frac{\pi}{2d} \left(\sum_{i=1}^{r} i \lfloor r/i \rfloor - r \right),$$

which is independent of ν. Also, every factor involved is nonzero. Therefore all numbers $h_\nu(\zeta_{2d})$ lie on a single open ray from the origin.

Since $r < d$, no factor $1 + x^i$ with $1 \le i \le r$ is divisible by Φ_{2d}. Thus $G(r, \zeta_{2d}) \neq 0$. Dividing by this fixed nonzero number only rotates and rescales the ray. Hence the reduced summands $q_\nu(\zeta_{2d})$ also lie on a single open ray. Their sum cannot be zero. □

Proof of Theorem 1. Every irreducible factor of $\operatorname{den}(n,x)$ is some $\Phi_{2d}(x)$ with $1 \le d \le n$. Fix such a d. By Lemma 4 and Lemma 5,

$$\operatorname{num}(n, \zeta_{2d}) \neq 0.$$

Hence $\Phi_{2d} \nmid \operatorname{num}(n,x)$. Therefore no nonconstant irreducible factor of $\operatorname{den}(n,x)$ divides $\operatorname{num}(n,x)$. Since $\operatorname{den}(n,x)$ has constant term 1, no rational integer prime divides all its coefficients. Thus $\gcd(\operatorname{num}(n,x), \operatorname{den}(n,x)) = 1$ in $\mathbb{Z}[x]$. □

4. Binary partitions: proof of Conjecture 5 and Conjecture 7

We next adapt the same root-of-unity idea to binary partitions. Here the allowed parts are $1, 2, 4, \ldots$, and the factors $1 + x^{2^k}$ are already pairwise coprime cyclotomic polynomials. By the definitions in the introduction,

$$h_{\mathcal{B},\lambda}(x) := \prod_{k \ge 0} \left(1 + x^{2^k}\right)^{\lfloor n/2^k \rfloor - m_\lambda(2^k)}.$$

In this context we assert that:

Lemma 6. *We always have $G_{\mathcal{B}}(n,x)=1$, for every n. Hence,*

$$\mathrm{num}_{\mathcal{B}}(n,x)=\sum_{\lambda\in\mathcal{B}(n)}h_{\mathcal{B},\lambda}(x)\qquad\text{and}\qquad \mathrm{den}_{\mathcal{B}}(n,x)=\prod_{k\geq 0}(1+x^{2^k})^{\lfloor n/2^k\rfloor}.$$

Proof. Since $1+x^{2^k}=\Phi_{2^{k+1}}(x)$, these factors are pairwise coprime irreducibles. For a fixed k, put $D:=2^k$. The binary partition with $\lfloor n/D\rfloor$ parts equal to D, together with any binary partition of the remainder $n-D\lfloor n/D\rfloor$, makes the exponent of $1+x^D$ equal to zero. Thus, no $1+x^{2^k}$ divides all binary summands. □

The same pairwise coprimality shows that Conjecture 5 is equivalent to Conjecture 7: the binary numerator is coprime to the binary denominator exactly when no factor $1+x^{2^s}$ divides $\mathrm{num}_{\mathcal{B}}(n,x)$.

Proof of Theorem 2. Fix $D:=2^s\leq n$, and let

$$\zeta:=e^{\pi \mathrm{i}/D}.$$

Then $1+x^D$ is the minimal polynomial of ζ, since D is a power of 2. It is enough to show that $\mathrm{num}_{\mathcal{B}}(n,\zeta)\neq 0$.

Write $n=aD+r$, with $a=\lfloor n/D\rfloor$ and $0\leq r<D$. In a binary summand $h_{\mathcal{B},\lambda}(x)$, the only factor $1+x^{2^k}$ that vanishes at $x=\zeta$ is $1+x^D$. Hence a summand can be nonzero only if

$$m_\lambda(D)=a.$$

The surviving partitions are exactly

$$\lambda=(D^a,\rho),\qquad \rho\in\mathcal{B}(r).$$

For such λ, comparison of the exponents gives the exact identity

$$h_{\mathcal{B},(D^a,\rho)}(x)=C_{n,D}(x)h_{\mathcal{B},\rho}(x),$$

where

$$C_{n,D}(x):=\prod_{\substack{k\geq 0\\ 2^k\neq D}}(1+x^{2^k})^{\lfloor n/2^k\rfloor-\lfloor r/2^k\rfloor}.$$

This product is independent of ρ. Since $1+\zeta^{2^k}\neq 0$ for $2^k\neq D$, we have $C_{n,D}(\zeta)\neq 0$. Therefore, we have

$$\mathrm{num}_{\mathcal{B}}(n,\zeta)=C_{n,D}(\zeta)\,\mathrm{num}_{\mathcal{B}}(r,\zeta).$$

It remains to prove $\mathrm{num}_{\mathcal{B}}(r,\zeta)\neq 0$. If $r=0$, this is immediate. Assume $1\leq r<D$. For every binary part $2^k\leq r$,

$$1+\zeta^{2^k}=2\cos\left(\frac{\pi 2^k}{2D}\right)e^{\pi \mathrm{i}2^k/(2D)},$$

and the cosine is positive. Thus each such factor has positive length and argument $\pi 2^k/(2D)$. For $\rho\in\mathcal{B}(r)$, the argument of $h_{\mathcal{B},\rho}(\zeta)$ is

$$\frac{\pi}{2D}\sum_{2^k\leq r}2^k\cdot\left(\left\lfloor r/2^k\right\rfloor-m_\rho(2^k)\right).$$

Since $\sum_k 2^k m_\rho(2^k)=r$, this argument is independent of ρ. All surviving terms are nonzero and lie on a single open ray. Their sum is nonzero. Therefore $\mathrm{num}_{\mathcal{B}}(n,\zeta)\neq 0$, so $1+x^D\nmid\mathrm{num}_{\mathcal{B}}(n,x)$. This proves Conjecture 7, and the coprimality statement follows from the equivalence above. □

5. Odd partitions: proof of Conjecture 8

We now specialize the restricted numerator for odd partitions at $x = -1$. The definitions from the introduction give

$$h_{\mathcal{O},\lambda}(x) \coloneqq \prod_{\substack{i\geq 1\\ i \text{ odd}}} (1+x^i)^{\lfloor n/i\rfloor - m_\lambda(i)},$$

with common divisor

$$G_{\mathcal{O}}(n,x) \coloneqq \gcd_{\lambda\in\mathcal{O}(n)} h_{\mathcal{O},\lambda}(x)$$

and reduced numerator

$$\operatorname{num}_{\mathcal{O}}(n,x) \coloneqq \frac{1}{G_{\mathcal{O}}(n,x)} \sum_{\lambda\in\mathcal{O}(n)} h_{\mathcal{O},\lambda}(x) = \sum_{\lambda\in\mathcal{O}(n)} q_{\mathcal{O},\lambda}(x)$$

where the reduced summands $q_{\mathcal{O},\lambda} \coloneqq \frac{h_{\mathcal{O},\lambda}}{G_{\mathcal{O},\lambda}}$ are defined in the obvious way. The proof shows that, after this cancellation, only the all-ones partition survives at $x = -1$.

Proof of Theorem 3. For an odd positive integer e, the factor Φ_{2e} divides $1 + x^i$, with i odd, exactly when $i = ej$ for an odd integer j. Therefore the exponent of Φ_{2e} in $h_{\mathcal{O},\lambda}$ is

$$\sum_{\substack{j\geq 1\\ j \text{ odd}}} \left(\left\lfloor \frac{n}{ej} \right\rfloor - m_\lambda(ej) \right).$$

As before, this is minimized by maximizing the number of parts of the form ej with j odd. In symbols, we have $\sum_{j\geq 1 \text{ odd}} m_\lambda(ej) \leq \lfloor n/e \rfloor$ with equality when λ has $\lfloor n/e \rfloor$ parts equal to e. Thus the exponent of Φ_{2e} in $G_{\mathcal{O}}(n,x)$ is

$$\sum_{\substack{j>1\\ j \text{ odd}}} \left\lfloor \frac{n}{ej} \right\rfloor.$$

Consequently, in $q_{\mathcal{O},\lambda}$, the exponent of Φ_{2e} is

$$\left\lfloor \frac{n}{e} \right\rfloor - \sum_{\substack{j\geq 1\\ j \text{ odd}}} m_\lambda(ej).$$

We now specialize to $x = -1$. The only cyclotomic factor that vanishes is $\Phi_2(x) = x + 1$, which corresponds to $e = 1$. The exponent of Φ_2 in the reduced summand is

$$n - \ell(\lambda),$$

where $\ell(\lambda)$ is the number of parts of λ. Among odd partitions of n, the maximum possible length is n, attained uniquely by (1^n). Hence all reduced summands vanish at -1 except the one corresponding to (1^n), that is to say,

$$\operatorname{num}_{\mathcal{O}}(n,-1) = q_{\mathcal{O},(1^n)}(-1) = \prod_{\substack{e>1\\ e \text{ odd}}} \Phi_{2e}(-1)^{\lfloor n/e\rfloor}.$$

By Lemma 2, $\Phi_{2e}(-1) = p$ if $e = p^a$ is a power of an odd prime p, and $\Phi_{2e}(-1) = 1$ otherwise. Therefore

$$\operatorname{num}_{\mathcal{O}}(n,-1) = \prod_{p \text{ odd prime}} \prod_{a\geq 1} p^{\lfloor n/p^a\rfloor} = o(n!),$$

by Legendre's formula with the prime 2 omitted. □

6. Ternary partitions: proofs of Conjecture 9 and Conjecture 10

Finally, we treat ternary partitions. The same cancellation analysis gives both the evaluation at $x=-1$ and the generating function needed at $x=1$. We use

$$h_{\mathcal{T},\lambda}(x) \coloneqq \prod_{k\geq 0}(1+x^{3^k})^{\lfloor n/3^k\rfloor - m_\lambda(3^k)},$$

$$G_{\mathcal{T}}(n,x) \coloneqq \gcd_{\lambda\in\mathcal{T}(n)} h_{\mathcal{T},\lambda}(x), \qquad \mathrm{num}_{\mathcal{T}}(n,x) \coloneqq \frac{1}{G_{\mathcal{T}}(n,x)}\sum_{\lambda\in\mathcal{T}(n)} h_{\mathcal{T},\lambda}(x) = \sum_{\lambda\in\mathcal{T}(n)} q_{\mathcal{T},\lambda}(x).$$

For every $k\geq 0$, we have

$$1+x^{3^k} = \prod_{a=0}^{k}\Phi_{2\cdot 3^a}(x).$$

Therefore, the exponent of $\Phi_{2\cdot 3^a}$ in $h_{\mathcal{T},\lambda}$ is

$$\sum_{k\geq a}\left(\left\lfloor\frac{n}{3^k}\right\rfloor - m_\lambda(3^k)\right).$$

As in the previous sections, this is minimized by using $\lfloor n/3^a\rfloor$ parts equal to 3^a and filling the remainder with ones. Hence, the exponent of $\Phi_{2\cdot 3^a}$ in $G_{\mathcal{T}}(n,x)$ is

$$\sum_{k>a}\left\lfloor\frac{n}{3^k}\right\rfloor,$$

and the exponent in a reduced summand $q_{\mathcal{T},\lambda}$ is

$$\left\lfloor\frac{n}{3^a}\right\rfloor - \sum_{k\geq a} m_\lambda(3^k).$$

Proof of Theorem 4. At $x=-1$, the only vanishing factor is $\Phi_2 = x+1$, corresponding to $a=0$. Its exponent in the reduced summand $q_{\mathcal{T},\lambda}$ is

$$n-\ell(\lambda).$$

The unique ternary partition of n with n parts is (1^n). Therefore only the reduced summand $q_{\mathcal{T},(1^n)}$ survives at -1.

For this summand, the reduced exponent of $\Phi_{2\cdot 3^a}$ is 0 for $a=0$, and is $\lfloor n/3^a\rfloor$ for $a\geq 1$. Therefore, we have

$$\mathrm{num}_{\mathcal{T}}(n,-1) = q_{\mathcal{T},(1^n)}(-1) = \prod_{a\geq 1}\Phi_{2\cdot 3^a}(-1)^{\lfloor n/3^a\rfloor}.$$

By Lemma 2, $\Phi_{2\cdot 3^a}(-1)=3$. Therefore, we have

$$\mathrm{num}_{\mathcal{T}}(n,-1) = 3^{\sum_{a\geq 1}\lfloor n/3^a\rfloor} = 3^{v_3(n!)}.$$

In view of

$$v_3((3n)!) = v_3((3n+1)!) = v_3((3n+2)!),$$

the block form in Conjecture 9 follows. □

Proof of Theorem 5. At $x=1$, Lemma 2 gives $\Phi_{2\cdot 3^a}(1)=1$ for $a\geq 1$. Thus the value of $G_{\mathcal{T}}(n,x)$ at $x=1$ is determined only by the factor $\Phi_2(1)=2$. Its exponent in $G_{\mathcal{T}}(n,x)$ is

$$\sum_{k\geq 1}\left\lfloor\frac{n}{3^k}\right\rfloor.$$

For a ternary partition $\lambda \in \mathcal{T}(n)$, we have

$$h_{\mathcal{T},\lambda}(1) = 2^{\sum_{k\geq 0}\lfloor n/3^k \rfloor - \ell(\lambda)}.$$

After division by $G_{\mathcal{T}}(n,1) = 2^{\sum_{k\geq 1}\lfloor n/3^k \rfloor}$, the reduced summand has value

$$q_{\mathcal{T},\lambda}(1) = 2^{n-\ell(\lambda)}.$$

Therefore, we obtain

$$t(n) = \mathrm{num}_{\mathcal{T}}(n,1) = \sum_{\lambda\in\mathcal{T}(n)} 2^{n-\ell(\lambda)}.$$

This has the ordinary generating function

$$T(q) := \sum_{n\geq 0} t(n)q^n = \prod_{j\geq 0}\left(1 - 2^{3^j-1}q^{3^j}\right)^{-1}.$$

The factor with $j = 0$ is $(1-q)^{-1}$. Reindexing the remaining product, we find

$$\prod_{j\geq 1}\left(1 - 2^{3^j-1}q^{3^j}\right)^{-1} = \prod_{j\geq 0}\left(1 - 2^{3^j-1}(4q^3)^{3^j}\right)^{-1} = T(4q^3).$$

Thus, we find that

$$T(q) = \frac{1}{1-q}T(4q^3), \qquad \text{or} \qquad (1-q)T(q) = T(4q^3).$$

Comparing coefficients of q^m, with the convention $t(-1) = 0$, gives

$$t(m) - t(m-1) = 0$$

unless m is divisible by 3. This proves

$$t(3n) = t(3n+1) = t(3n+2) \qquad (n \geq 0).$$

Comparing coefficients of q^{3n} gives

$$t(3n) - t(3n-1) = 4^n t(n).$$

By block constancy, $t(3n-1) = t(3n-2)$. Hence

$$t(3n) - t(3n-2) = 4^n t(n) = 2^{2n}t(n),$$

which is the claimed finite-difference formula. □

7. Appendix: Autonomous proof production and formalization

At Axiom Math, we are developing AxiomProver, an AI system for mathematical research based on autoformalization. As a test case, we gave AxiomProver ten tasks: to autonomously formalize and prove Conjectures 1-10, offering examples of AI assistance as well as limitations in mathematical research. This appendix is separate from the rest of the manuscript.

7.1. **Division of labor.** The prose exposition of this paper, including this appendix, was written without the use of AI. AxiomProver runs Lean, an interactive theorem prover and a functional programming language built on dependent type theory, designed to provide a rigorous computational framework for validating mathematical proofs [4]. This appendix is included not only to record that formal proofs were produced, but also to clarify the division of labor between the human-written mathematics and the AI-generated formalizations. The mathematical statements supplied to AxiomProver were written in natural language and can be found in [2].

The role of AxiomProver was to convert the non-formal statements into Lean and to construct machine-checkable formal proofs. Thus, in the tasks described below, AxiomProver was neither being asked to make any of the conjectures, nor to decide the final organization of the paper; it was being asked to produce complete formal proofs of the supplied statements.

Autoformalization involves automatically and autonomously converting natural-language mathematics into machine-verifiable formal language. Lean files are created to pass type checkers, while the natural language papers aim to communicate ideas to readers.

7.2. **Task specification and results.** Given problems in natural or formal language, AxiomProver attempts to generate a complete formal proof. When it succeeds, the system outputs two files:

- `problem.lean`, which formalizes the problem statement;
- `solution.lean`, which represents a complete proof in a formal language.

The ten tasks share a uniform layout — for $i \in \{1, \dots, 10\}$, we gave the following files as input for the i-th task:

- `Conj_`i`_Statement.tex`
- `task.md` contains a single line

  Please read '`Conj_`i`_Statement.tex`' and formalize Conjecture `\ref{conj:conj_`i`}`.
- `.environment` specifies the Lean version used is 4.28.0.

Out of the 10 tasks, AxiomProver was able to autonomously formalize and prove Conjecture j for $j \in \{2, 5, 7, 8, 9, 10\}$. It also found the counterexample $n = 4$ to the part of [2, Conjecture 6] asserting that $\mathrm{num}_{\mathcal{B}}(n, x)$ has log-concave coefficients (see the remark at the end of Section 1.2.2). Here "autonomously" means that, after receiving the specified input files, AxiomProver generated the corresponding `problem.lean` and `solution.lean` files without further mathematical intervention in the proof script. The authors did not hand-write or repair the Lean proofs line by line. The resulting formal proofs were then used as machine-checkable certificates for the proved conjectures.

7.3. **Artifacts.** The relevant files are posted in the following repository:

https://github.com/AxiomMath/PartitionPolynomial

The version of Lean used is 4.28.0. Compatibility with other versions is not guaranteed due to the evolving nature of the Lean 4 compiler and its core libraries.

At first glance, the proofs generated by AxiomProver do not resemble the narrative outlined in this paper. Converting a Lean file into a proof understandable by humans is challenging because Lean is designed as code for a type-checker, not as a reader-friendly explanation. It makes all the "obvious" bookkeeping explicit, such as rewrite steps, coercions, side conditions, and case splits, and tends to follow the most convenient lemmas and tactics for the library, rather than the most clear conceptual route. A mathematician can usually condense this significantly by relying on shared historical context, standard arguments, and informal identifications that Lean cannot assume. As a result, writing a paper from Lean files is not just about reformatting. The authors must understand the formal script, reconstruct the underlying ideas, and then translate the code into a narrative that emphasizes the key insights while safely omitting routine details.

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